

Linson Philip

Electrical & Energy Data Engineer

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Professional Profile

Electrical and energy-data engineer with experience in EHV power-system operations, energy-meter testing, measurement-data analytics, digital-twin development and laboratory automation. Currently working on data-driven assessment of electricity meters and automated three-phase test systems. Hands-on experience with Python, SQL, Node-RED, Modbus, RS-485, GPIB/VISA, MATLAB/Simulink and industrial measurement equipment.

Core Competencies

Power Systems	Power systems, energy-meter testing, accuracy assessment, electrical measurements, three-phase systems, power quality
Data Analysis	Python, SQL, Pandas, NumPy, data validation, KPI development, statistical and time-series analysis
Lab Automation	Node-RED, automated test sequences, data acquisition, equipment monitoring, communication handling
Communication	Modbus TCP/RTU, RS-485, GPIB, VISA, Ethernet-based laboratory integration
Modelling	MATLAB, Simulink, Simscape, LTspice, digital twins, parameter identification
Databases	Microsoft SQL Server, MySQL, InfluxDB

Professional Experience

Academic Research Associate / Research Engineer <i>Institute for Sustainable Energy Systems (INES), Hochschule Offenburg</i>	05/2025–Present <i>Offenburg, Germany</i>
- Analyse more than two million historical energy-meter test records using Python, Pandas and SQL Server to evaluate measurement error, long-term behaviour and device-level performance. - Develop data-processing workflows for test-point harmonisation, measurement validation, KPI generation, repeated-meter analysis and automated technical reporting. - Design and integrate an automated three-phase energy-meter test laboratory with programmable power equipment, electronic loads, reference measurement systems, tariff meters and climatic test equipment. - Develop equipment-control and data-acquisition workflows using Python, Node-RED, Modbus TCP/RTU, RS-485 and GPIB/VISA, with structured measurement storage in MySQL and InfluxDB. - Define reproducible test sequences across different current levels, phase configurations and power-factor conditions in collaboration with academic and industrial project partners.	
Master's Thesis Student – Digital Twin Development <i>Livarsa GmbH & Hochschule Offenburg</i>	09/2024–02/2025 <i>Germany</i>
- Developed a measurement-based digital twin to assess the operating behaviour and energy-saving potential of an industrial passive electrical efficiency filter. - Conducted electrical measurements and parameter identification to derive model parameters and validate simulation behaviour. - Built and validated LTspice and MATLAB/Simscape models using real measurement data and compared simulated and measured system responses.	
Student Research Assistant – Micro-PV Monitoring <i>Institute for Sustainable Energy Systems, Hochschule Offenburg</i>	12/2023–09/2024 <i>Offenburg, Germany</i>
- Supported installation, monitoring and maintenance of micro-photovoltaic systems and collected operational energy data. - Developed Python-based workflows for data preparation, system-performance analysis and technical reporting.	

- Operated and monitored 11–110 kV substations and transmission equipment to support safe and reliable electricity supply.
- Performed switching operations, equipment inspections, insulation-resistance testing and operational documentation.
- Worked with power transformers, circuit breakers, isolators, protection relays, control panels, battery systems and auxiliary supplies.
- Monitored voltage, current, loading and energy import/export conditions and supported maintenance, shutdown and restoration activities.

Selected Engineering Projects

Data-Driven Accuracy Assessment of Energy Meters

2025–Present

Development of a reproducible workflow for historical meter-test data covering test-point harmonisation, measurement-error analysis, long-term stability assessment, device-level KPIs and automated reporting.

Automated Three-Phase Energy-Meter Test Laboratory

2026–Present

Integration of programmable power equipment, electronic loads, reference measurements, energy meters and databases through Python, Node-RED, Modbus TCP/RTU, RS-485 and GPIB/VISA.

Digital Twin for Electrical Efficiency Filter Assessment

2024–2025

Measurement-based parameter identification and validation of simulation models for pre-installation assessment of an industrial electrical efficiency system.

Education

M.Sc. Renewable Energy and Data Engineering

2023–2025

Hochschule Offenburg

Germany

Master’s thesis: Digital-twin development for performance and energy-saving assessment of an industrial electrical efficiency system.

B.Tech. Electrical and Electronics Engineering

2014–2018

University of Kerala

India

Technical Skills

Programming	Python, SQL, Pandas, NumPy, Scikit-learn, statistical analysis, time-series analysis
Automation	Node-RED, Modbus TCP/RTU, RS-485, GPIB, VISA, data acquisition
Engineering Tools	MATLAB, Simulink, Simscape, LTspice
Measurement	Energy-meter testing, electrical measurements, power systems, EHV substations, transformers, switchgear, protection systems
Databases	Microsoft SQL Server, MySQL, InfluxDB

Training

Python for Data Science (DataCamp)

Languages & Additional Information

Languages	Malayalam (Native), English (C1), German (B2), Tamil (B1), Hindi (B1)
Driving Licence	German driving licence, Class B197
Location	Ortenberg, Germany